

# Rowan Torbitzky-Lane

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## Education

B.S. Mathematics - GPA: 4.00 Lindenwood University

Expected: May 2027

B.S. Computer Science - GPA: 3.80 Missouri University of Science and Technology

Graduated: December 2024

## Skills

- |          |         |         |
|----------|---------|---------|
| • Python | • Unity | • CUDA  |
| • Rust   | • SQL   | • Godot |
| • Docker | • Linux | • Nix   |

## Relevant Coursework

- |                      |                              |                       |
|----------------------|------------------------------|-----------------------|
| • Algorithms         | • Differential Equations     | • Linear Algebra I/II |
| • Data Structures    | • Calculus III               | • Physics I/II        |
| • Introduction to AI | • Probability and Statistics | • Linear Regression   |

## Experience

**Computer Vision Intern** - *USC Institute for Creative Technologies*

May 2026 - August 2026

- Accelerate mesh/splat correspondence calculation to under one second via C# to Rust FFI
- Train Gaussian Splats on bounded and unbounded scenes
- Extract single objects from Gaussian Splats
- Utilize Nix for programming environment setup

**NSF Research Assistant** - *Missouri University of Science and Technology*

May 2024 - October 2024

- Train novel, bi-directional LLMs on Genome Sequences to correct errors in reads
- Clean and analyze textual data with Scikit-learn and Scikit-bio
- Incorporate huggingface library for model testing

**Research Assistant** - *Center for Intelligent Infrastructure*

May 2023 - October 2023

- Calculate optical flow with MATLAB and C++ on campus HPC via Slurm
- Coordinate Web Requests with Django to interact with internal PostgreSQL server
- Improve frontend with Typescript, such as adding video upload capability
- Extract image features with OpenCV, using optical flow to predict annotations in subsequent video frames
- Perform Annotations for image segmentation Convolutional Neural Networks

**Data Science Co-Op** - *Hunter Engineering Company*

January 2022 - August 2022

- Utilize YOLO and PyTorch in license plate recognition problems
- Pioneer object detection models to eradicate bad data before annotators see it
- Innovate data pipeline by automating our data downloading process
- Design and containerize Python APIs with FastAPI and Flask within Docker and begin creating endpoints within Azure
- Leverage Pandas and Microsoft SQL in data processing pipelines
- Enforce consistent code style with automated formatting checks in internal Azure DevOps pipeline

**Introduction to CS Grader** - *Missouri University of Science and Technology*

January 2021 - May 2021

- Author automated Python test cases validating student submissions against assignment specifications
- Translate test failures into actionable feedback for students new to programming

## Personal Projects

**Self-Hosted NixOS Server** - *SQL, YOLO, Docker, FirewallD, Git, Nginx, CUDA*

January 2023 - Present

- Automate flake updates daily via CI/CD pipeline on GitLab
- Train bird object detection Convolutional Neural Networks
- Host game servers via Docker

**Rust Genetic Programming Framework** - *Polars, rust\_decimal*

March 2025 - Present

- Perform pre-processing on Time Series Data
- Utilize test driven development philosophy involving writing unit tests
- Write declarative and procedural macros for reduced boilerplate
- Utilize rust\_decimal's floating point precision for increased accuracy

## Talks & Events

*Talk: Genetic Programming in Python @ PySTL* [https://www.youtube.com/watch?v=ZzYi\\_j2IYuY](https://www.youtube.com/watch?v=ZzYi_j2IYuY)

September 18th, 2025

*Talk: Actix Web Framework @ STL Rust* <https://rumble.com/v72mid6-actix-web-unleashed.html>

December 4th, 2025

*Hackathon: SIGGRAPH 2026 Worlds in Action Hack @ MG Studio* [luma.com/u7pmhw92](https://luma.com/u7pmhw92)

July 18th, 2026